Lesson 15: Neyman-Pearson Lemma; Likelihood Ratio Tests

BSTA 551: Statistical Inference

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Review: Where We Left Off

Lesson 14: P-Values; Power and Sample Size Determination

- P-values are Uniform(0,1) under H_0 by the probability integral transformation
- Power depends on the noncentrality parameter $\delta = (\mu' - \mu_0)/(\sigma/\sqrt{n})$
- Sample size determination via `power.t.test()` for planning clinical trials
- Multiple testing adjustments: Bonferroni (FWER) and Benjamini-Hochberg (FDR)

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Today's Goals:

1. Understand the concept of a **most powerful** test for simple hypotheses
2. State and prove the **Neyman-Pearson Lemma**
3. Define **power functions** and **uniformly most powerful (UMP)** tests
4. Develop the **likelihood ratio test (LRT)** for composite hypotheses
5. Connect the LRT to familiar tests (z test, t test)
6. Apply the large-sample chi-squared approximation for LRTs

Roadmap for Today

Topic	Key Idea	Textbook
Simple hypotheses	Both H_0 and H_a fully specify the distribution	Devore 9.5
Neyman-Pearson Lemma	The likelihood ratio test is most powerful for simple H_0 vs simple H_a	Devore 9.5
Power function	$\pi(\theta') = P_{\theta'}(\text{Reject } H_0)$ as a function of θ	Devore 9.5
UMP tests	Most powerful for every value in the alternative	Devore 9.5
Likelihood ratio tests	General principle for composite hypotheses	Devore 9.5
LRT $\to$ t test	The one-sample t test is a LRT	Devore 9.5
Large-sample LRT	$-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$	Devore 9.5

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Why This Matters

The Neyman-Pearson Lemma provides the **theoretical foundation** for optimal hypothesis testing. It tells us *why* the tests we have been using (z tests, t tests) are the best possible tests in their respective settings. The likelihood ratio principle then gives us a **general recipe** for constructing tests in new situations.

Simple Hypotheses and Optimal Tests (Devore 9.5)

Simple vs. Composite Hypotheses

Recall from Lesson 12:

Definition (Simple and Composite Hypotheses)

A hypothesis is **simple** if it completely specifies the distribution of the sample $X_1, \ldots, X_n$. Otherwise, it is **composite**.

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- $X_1, \ldots, X_n \overset{iid}{\sim} \text{Exp}(\lambda)$:
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- $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **known**:
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  - $H : \mu \neq 100$ is **composite**
- $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**:
  - $H : \mu = 100$ is **composite** (because σ is unspecified — it is a *nuisance parameter*)

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For a test of $H_0 : \theta = \theta_0$ vs $H_a : \theta = \theta_a$, both simple, there are infinitely many possible rejection regions R. Each one gives:

- A **Type I error rate**: $\alpha = P_{\theta_0}((X_1, \ldots, X_n) \in R)$
- A **Type II error rate**: $\beta = P_{\theta_a}((X_1, \ldots, X_n) \notin R)$

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Goal: Among all tests with $\alpha \leq \alpha^*$ (controlling Type I error), find the test that **minimizes** β (equivalently, **maximizes power**).

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Medical Research Motivation

In a clinical trial comparing a new drug to a standard, we want the test that gives us the **best chance of detecting a real treatment effect** (high power) while maintaining the required significance level. The Neyman-Pearson Lemma tells us exactly how to construct such a test.

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- If $\Lambda^* \gg 1$: data much more consistent with $H_a \Rightarrow$ evidence against H_0
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Convention Note

Devore defines the NP ratio as $L(\theta_a)/L(\theta_0)$ (reject when large). Chihara defines it as $L(\theta_0)/L(\theta_a)$ (reject when small). Both formulations are equivalent — just inverted. We will primarily follow Devore's convention in the statement of the lemma.

The Neyman-Pearson Lemma

Statement of the Neyman-Pearson Lemma

The Neyman-Pearson Lemma (Devore 9.5)

For testing a simple null hypothesis $H_0 : \theta = \theta_0$ versus a simple alternative $H_a : \theta = \theta_a$, let k be a fixed positive number and form the rejection region

$$R^* = \left\{(x_1, \ldots, x_n) : \frac{f(x_1, \ldots, x_n; \theta_a)}{f(x_1, \ldots, x_n; \theta_0)} \geq k\right\}$$

Let $\alpha^* = P((X_1, \ldots, X_n) \in R^* \text{ when } \theta = \theta_0)$ and let β^* denote the Type II error probability.

Then for **any other** test procedure with Type I error probability $\alpha \leq \alpha^*$, the Type II error probability must satisfy $\beta \geq \beta^*$.

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Then for **any other** test procedure with Type I error probability $\alpha \leq \alpha^*$, the Type II error probability must satisfy $\beta \geq \beta^*$.

In plain language: The test based on the likelihood ratio

- Has the **smallest** β (Type II error) among all tests with $\alpha \leq \alpha^*$
- Equivalently, has the **largest power** at θ_a among all level-α^* tests
- Is called the **most powerful (MP)** test at level α^*

Proof of the Neyman-Pearson Lemma

Let R be the rejection region of **any** test, so that:

$$\alpha = \sum_R f(x_1, \ldots, x_n; \theta_0), \quad \beta = 1 - \sum_R f(x_1, \ldots, x_n; \theta_a)$$

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For any constant $k > 0$, consider the linear combination $k\alpha + \beta$:

$$k\alpha + \beta = k \sum_R f(\mathbf{x}; \theta_0) + 1 - \sum_R f(\mathbf{x}; \theta_a) = 1 + \sum_R [k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)]$$

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Key insight: The expression $k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a)$ can be positive or negative. To **minimize** $k\alpha + \beta$, choose R to include exactly the points where this expression is negative:

$$R^* = \{\mathbf{x} : k \cdot f(\mathbf{x}; \theta_0) - f(\mathbf{x}; \theta_a) \leq 0\} = \left\{\mathbf{x} : \frac{f(\mathbf{x}; \theta_a)}{f(\mathbf{x}; \theta_0)} \geq k\right\}$$

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This is exactly the Neyman-Pearson rejection region! So R^* minimizes $k\alpha + \beta$.

Proof (continued)

We've shown: $k\alpha^* + \beta^* \leq k\alpha + \beta$ for **all** rejection regions R.

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What Does This Mean?

The proof works by showing that the NP rejection region simultaneously minimizes **both** α and β in a trade-off sense. You cannot do better on one without doing worse on the other. The NP test is on the **efficient frontier** of the (α, β) trade-off.

Example: Adverse Events in a Drug Trial (Normal, Known σ)

A pharmaceutical company monitors a biomarker level in patients. Under the current treatment, biomarker levels follow $N(\mu, 1)$. An elevated biomarker signals a potential adverse event.

Test $H_0 : \mu = 5$ vs $H_a : \mu = 6$ based on n patients (known $\sigma = 1$).

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Applying the Neyman-Pearson Lemma:

$$\frac{L(\mu_a = 6)}{L(\mu_0 = 5)} = \frac{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-(x_i-6)^2/2}}{\prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}} e^{-(x_i-5)^2/2}} = e^{\sum(x_i-6)^2/2\cdot(-1)+\sum(x_i-5)^2/2}$$

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Simplifying the exponent:

$$= e^{(6-5)\sum x_i - n(6^2-5^2)/2} = e^{\sum x_i - 11n/2}$$

Example: Normal (continued)

The rejection region $\Lambda^* \geq k$ becomes:

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Taking logarithms (since $\mu_a - \mu_0 = 1 > 0$, the exponential is increasing in $\sum x_i$):

$$\sum x_i \geq k' \quad \Longleftrightarrow \quad \bar{x} \geq k'' \quad \Longleftrightarrow \quad \frac{\bar{x} - 5}{1/\sqrt{n}} \geq c$$

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Key Result (Devore Example 9.24)

For $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ known, testing $H_0 : \mu = \mu_0$ vs $H_a : \mu = \mu_a$ where $\mu_a > \mu_0$:

The Neyman-Pearson Lemma shows that the **one-sample z test** (reject when $z \geq z_\alpha$) is the **most powerful** test at level α. Our familiar z test is optimal!

Example: Poisson Adverse Event Counts (Devore Example 9.23)

In post-market drug surveillance, the number of adverse events per reporting period is monitored. Suppose $X_1, \ldots, X_5 \overset{iid}{\sim} \text{Poisson}(\lambda)$ represent adverse event counts over 5 periods.

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The likelihood ratio:

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Taking logarithms: $\sum x_i \geq c$ where $c = \ln(k \cdot e^5) / \ln(2)$.

Example: Poisson (continued)

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If we use $c = 10$: $\alpha^* = P(Y \geq 10 \mid Y \sim \text{Poisson}(5)) = 1 - 0.968 = 0.032$

If we use $c = 9$: $\alpha^* = P(Y \geq 9 \mid Y \sim \text{Poisson}(5)) = 1 - 0.932 = 0.068$

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If we use $c = 9$: $\alpha^* = P(Y \geq 9 \mid Y \sim \text{Poisson}(5)) = 1 - 0.932 = 0.068$

For $\alpha \leq 0.05$, we use $c = 10$, giving $\alpha^* = 0.032$.

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Type II error: Under $H_a, Y \sim \text{Poisson}(10)$.

$$\beta^* = P(Y < 10 \mid Y \sim \text{Poisson}(10)) = 0.458$$

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▶ Code

```
alpha* = 0.03182806
```

▶ Code

```
beta*  = 0.4579297
```

▶ Code

```
Power  = 0.5420703
```

Example: Poisson (continued)

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Example: Poisson (continued)

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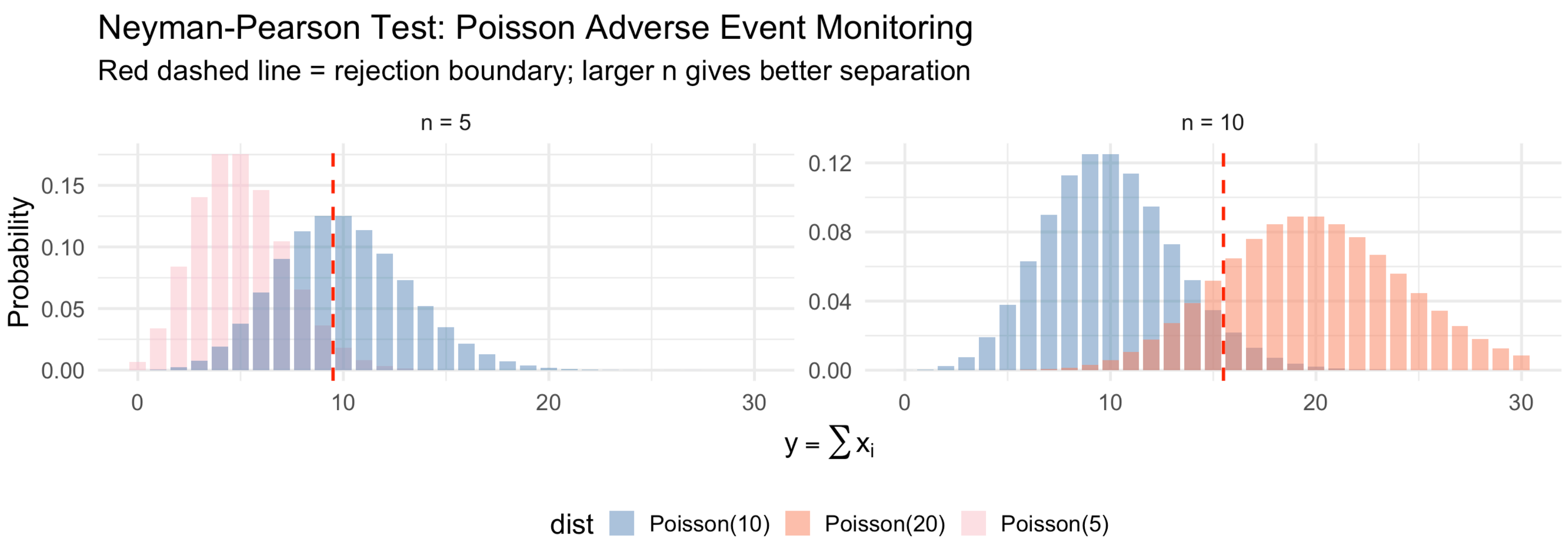

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Under H_0: $\sum_{i=1}^{9} X_i \sim \mathrm{Gamma}(9, 8)$. For $\alpha = 0.05$:

```r
c2 <- qgamma(0.05, shape = 9, rate = 8)
cat("Critical value c2 =", round(c2, 4), "\n")
```

```
Critical value c2 = 0.5869
```

By the Neyman-Pearson Lemma, this is the best possible test at this significance level.

Power Functions and UMP Tests

The Power Function

Definition (Power Function, Devore 9.5)

Let Ω_0 and Ω_a be two disjoint sets of possible values of θ. For testing $H_0 : \theta \in \Omega_0$ vs $H_a : \theta \in \Omega_a$ with rejection region R, the **power function** is:

$$\pi(\theta') = P_{\theta'}((X_1, \dots, X_n) \in R)$$

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Connection to error probabilities:

$$\pi(\theta') = \begin{cases} P(\text{Type I error}) = \alpha(\theta') & \text{when } \theta' \in \Omega_0 \\ 1 - P(\text{Type II error}) = 1 - \beta(\theta') & \text{when } \theta' \in \Omega_a \end{cases}$$

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Ideal power function (unachievable in practice):

$$\pi(\theta') = \begin{cases} 0 & \text{when } \theta' \in \Omega_0 \\ 1 & \text{when } \theta' \in \Omega_a \end{cases}$$

Visualizing the Power Function

Example (Devore 9.25 adapted): Blood pressure reduction (mmHg) with a new antihypertensive follows $N(\mu, 81)$. Test $H_0 : \mu \leq 75$ vs $H_a : \mu < 75$ with $n = 100, \alpha = 0.10$.

Rejection region: $z \leq -z_{0.10} = -1.28$, i.e., $\bar{x} \leq 73.848$.

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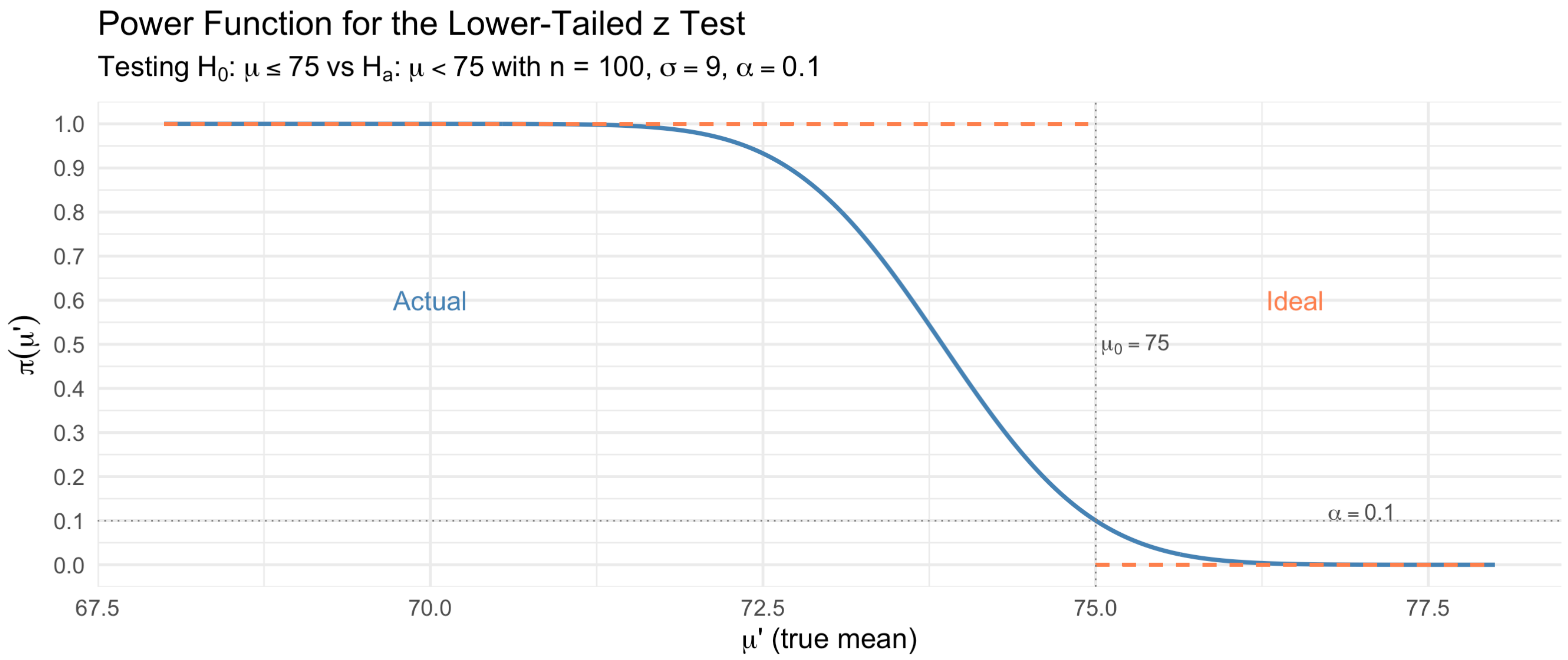

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Clinical Implication

In designing a clinical trial, we need to choose n large enough so that the power function is acceptably close to 1 at the **minimum clinically important difference**. This connects directly to the sample size calculations from Lesson 14.

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When do UMP tests exist? If the NP rejection region **does not depend on the specific alternative** θ_a, then the same test is most powerful for *every* $\theta_a \in \Omega_a$.

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Returning to the Poisson adverse event example with $n = 5$:

Test $H_0 : \lambda \leq 1$ vs $H_a : \lambda > 1$.

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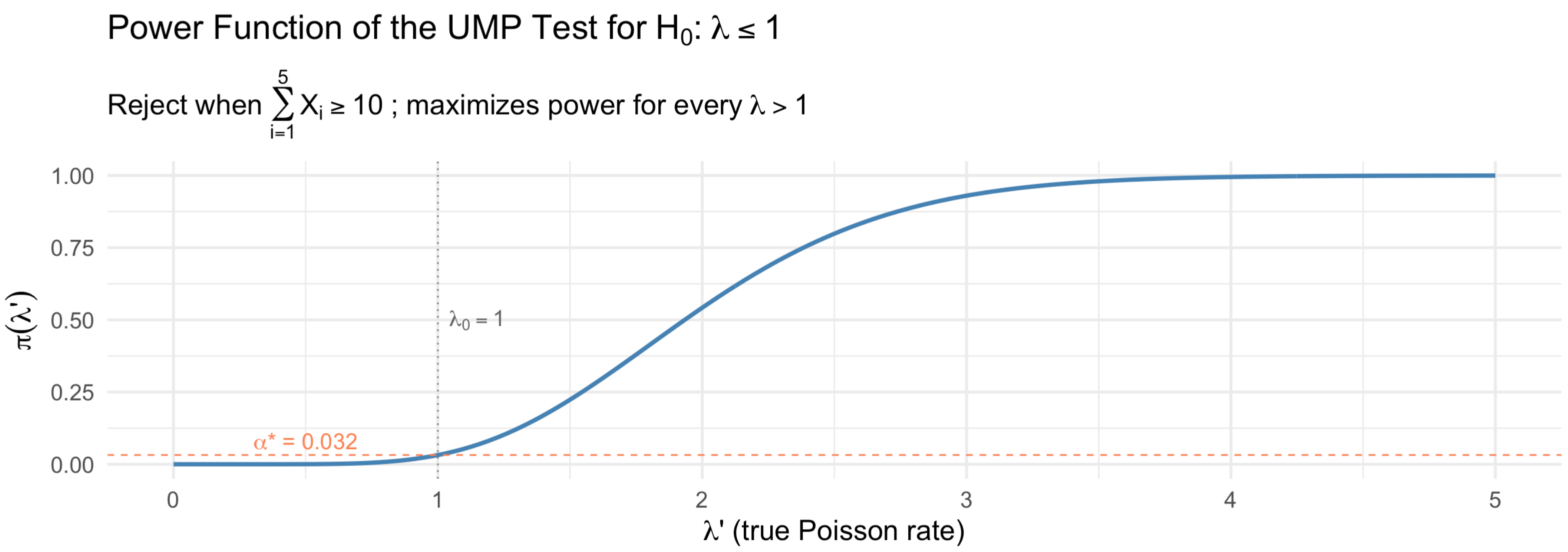

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Example: Testing $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ with $X_i \overset{iid}{\sim} N(\mu, \sigma^2)$, σ known.

- For $\mu_a > \mu_0$: NP test rejects when $\bar{x}$ is **large** (upper tail)
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Resolution: Restrict attention to **unbiased tests** (tests where $\pi(\theta') \geq \alpha$ for all $\theta' \in \Omega_a$). Under this restriction:

- The two-sided z test is **UMP unbiased** when σ is known
- The one-sample t test is **UMP unbiased** when σ is unknown

Unbiased Tests

Definition (Unbiased Test)

A test is **unbiased** if the power under any alternative is at least as large as the power under any null value:

$$\inf_{\theta' \in \Omega_a} \pi(\theta') \geq \sup_{\theta' \in \Omega_0} \pi(\theta')$$

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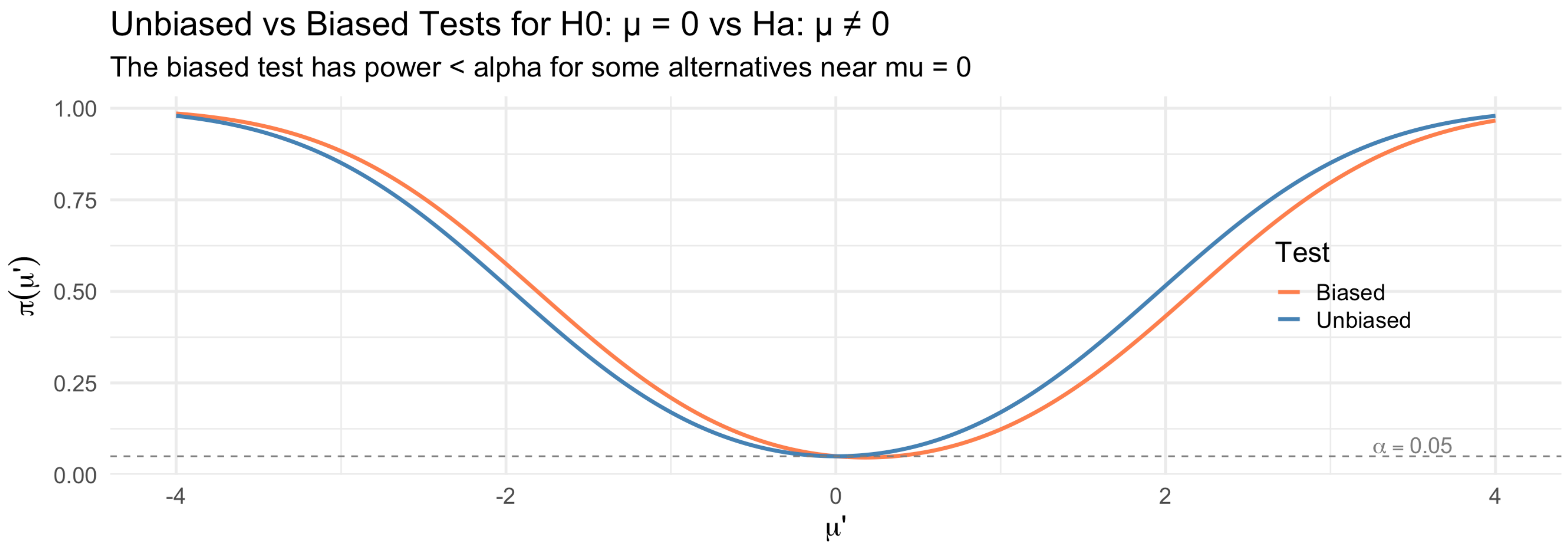

Likelihood Ratio Tests (Devore 9.5)

From Simple to Composite Hypotheses

The NP Lemma handles simple vs simple hypotheses. But in practice, we need tests for:

$$H_0 : \theta \in \Omega_0 \quad \text{vs} \quad H_a : \theta \in \Omega_a$$

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The likelihood ratio principle provides a **general recipe** for constructing test statistics in virtually any testing situation.

The Likelihood Ratio Test (LRT) Procedure

1. Find $\hat{\theta}_{mle}$, the MLE of θ over the **full** parameter space $\Omega = \Omega_0 \cup \Omega_a$
2. Find $\hat{\theta}_0$, the **restricted MLE** of θ over Ω_0 only
3. Form the **likelihood ratio statistic**:

$$\Lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{mle})}$$

4. Reject H_0 when $\Lambda \leq k$ (i.e., when the restricted MLE fits much worse than the unrestricted MLE)

Understanding the LRT Statistic

$$\Lambda = \frac{L(\hat{\theta}_0)}{L(\hat{\theta}_{mle})} = \frac{\max_{\theta \in \Omega_0} L(\theta)}{\max_{\theta \in \Omega} L(\theta)}$$

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Key properties:

- Always $0 \leq \Lambda \leq 1$ (because $\Omega_0 \subseteq \Omega$, so the restricted maximum $\leq$ unrestricted maximum)
- $\Lambda \approx 1$: the data are about equally consistent with H_0 and the overall best fit
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Connection to NP Lemma: When both hypotheses are simple ($\Omega_0 = \{\theta_0\}$ and $\Omega_a = \{\theta_a\}$):

$$\Lambda = \frac{L(\theta_0)}{\max\{L(\theta_0), L(\theta_a)\}} = \begin{cases} L(\theta_0)/L(\theta_a) & \text{if } L(\theta_a) \geq L(\theta_0) \\ 1 & \text{otherwise} \end{cases}$$

This is the reciprocal of the NP statistic (when H_a is more likely), so rejecting for small Λ is equivalent to rejecting for large Λ^*.

LRT Example 1: Normal Mean, Known Variance (Devore Example 9.27 style)

Systolic blood pressure readings follow $N(\mu, 100)$ (known $\sigma = 10$). A clinical trial tests whether a new treatment changes the population mean from a target value.

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Since $\sum(x_i - \mu_0)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \mu_0)^2$:

$$\Lambda = e^{-n(\bar{x}-\mu_0)^2/(2\sigma^2)}$$

LRT $\Rightarrow$ z Test

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Key Result

The **likelihood ratio test** for $H_0 : \mu = \mu_0$ vs $H_a : \mu \neq \mu_0$ when σ is known is exactly the **two-sided z test**. The LRT principle automatically “discovers” the z test.

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Now suppose $X_1, \ldots, X_n \overset{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**.

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$$\hat{\mu}_0 = \mu_0, \quad \hat{\sigma}^2_0 = \frac{1}{n} \sum (x_i - \mu_0)^2$$

LRT Example 2: Normal Mean, Unknown Variance (Devore Example 9.27)

Now suppose $X_1, \ldots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$ with σ **unknown**.

$$H_0 : \mu = \mu_0 \quad \text{vs} \quad H_a : \mu \neq \mu_0$$

Step 1: Unrestricted MLE over $\Omega = \{(\mu, \sigma^2) : \mu \in \mathbb{R}, \sigma^2 > 0\}$:

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Note: under H_0, we still estimate σ^2 by MLE, but with μ fixed at μ_0.

LRT $\Rightarrow$ t Test (Devore Example 9.27)

Step 3: Likelihood ratio. After substitution and simplification:

$$\Lambda = \left[\frac{\hat{\sigma}^2_{mle}}{\hat{\sigma}^2_0}\right]^{n/2} = \left[\frac{\sum(x_i - \bar{x})^2}{\sum(x_i - \mu_0)^2}\right]^{n/2}$$

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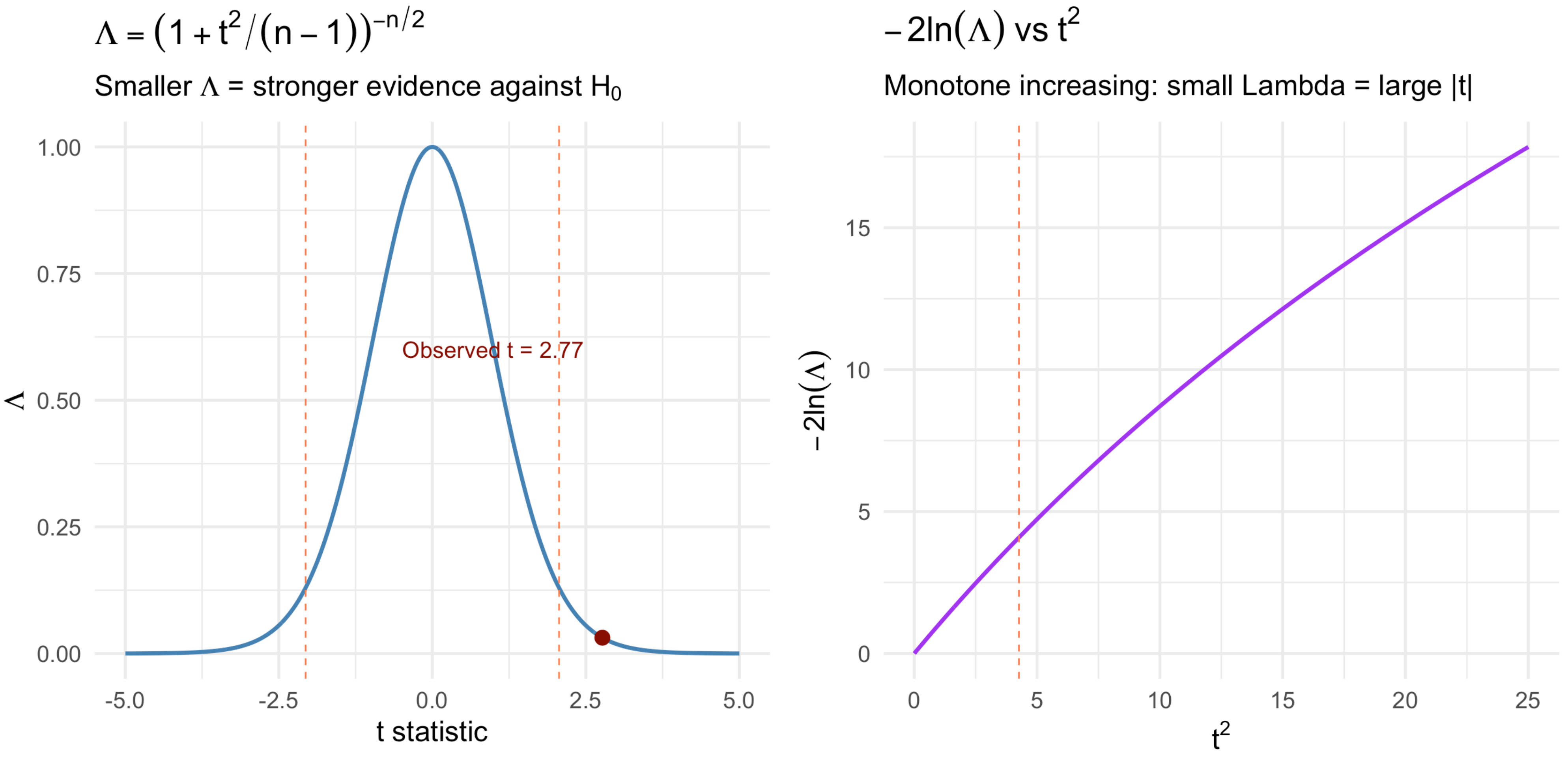
> **Key Result**
>
> The **likelihood ratio test** for $H_0: \mu = \mu_0$ vs $H_a: \mu \neq \mu_0$ when σ is unknown is exactly the **one-sample t test**. Many classical tests can be derived from the LRT principle.

LRT and the t Test: Simulation Illustration

▶ Code



Medical Application: Drug Efficacy Trial

A Phase III clinical trial tests whether a new cholesterol-lowering drug changes mean LDL from baseline $\mu_0 = 130$ mg/dL. Data from $n = 30$ patients:

```r
set.seed(42)
n <- 30
mu_0 <- 130
ldl_data <- rnorm(n, mean = 122, sd = 18)

x_bar <- mean(ldl_data)
s_val <- sd(ldl_data)
t_obs <- (x_bar - mu_0) / (s_val / sqrt(n))

cat("Sample mean:", round(x_bar, 2), "mg/dL\n")
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Sample mean: 123.23 mg/dL
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Large-Sample LRT Approximation

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Theorem (Wilks' Theorem, Devore 9.5)

If the sample size n is sufficiently large, then under H_0:

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The degrees of freedom m:

- $m = \dim(\Omega) - \dim(\Omega_0)$ = (number of parameters estimated under H_a) minus (number estimated under H_0)
- Represents the number of "constraints" imposed by H_0

Computing Degrees of Freedom: Examples

Setting	Ω	Ω_0	m
$N(\mu, \sigma^2)$, σ known, $H_0 : \mu = \mu_0$	1 (μ)	0	1
$N(\mu, \sigma^2)$, σ unknown, $H_0 : \mu = \mu_0$	2 (μ, σ^2)	1 (σ^2)	1
Bivariate normal, $H_0 : \mu_1 = \mu_2$ and $\sigma_1 = \sigma_2$	5 ($\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$)	3 (μ, σ, ρ)	2
Multinomial with k categories, $H_0 : p_i = p_{i,0}$	$k - 1$	0	$k - 1$

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Practical Rule

Count the number of parameters that are “free” (need to be estimated) under each hypothesis. The difference is m.

Large-Sample LRT Example: Double Exponential (Devore 9.5)

The Laplace (double exponential) distribution with pdf $f(x;\theta) = \frac{1}{2}e^{-|x-\theta|}$ is sometimes used to model measurement errors. Test $H_0 : \theta = 0$ vs $H_a : \theta \neq 0$.

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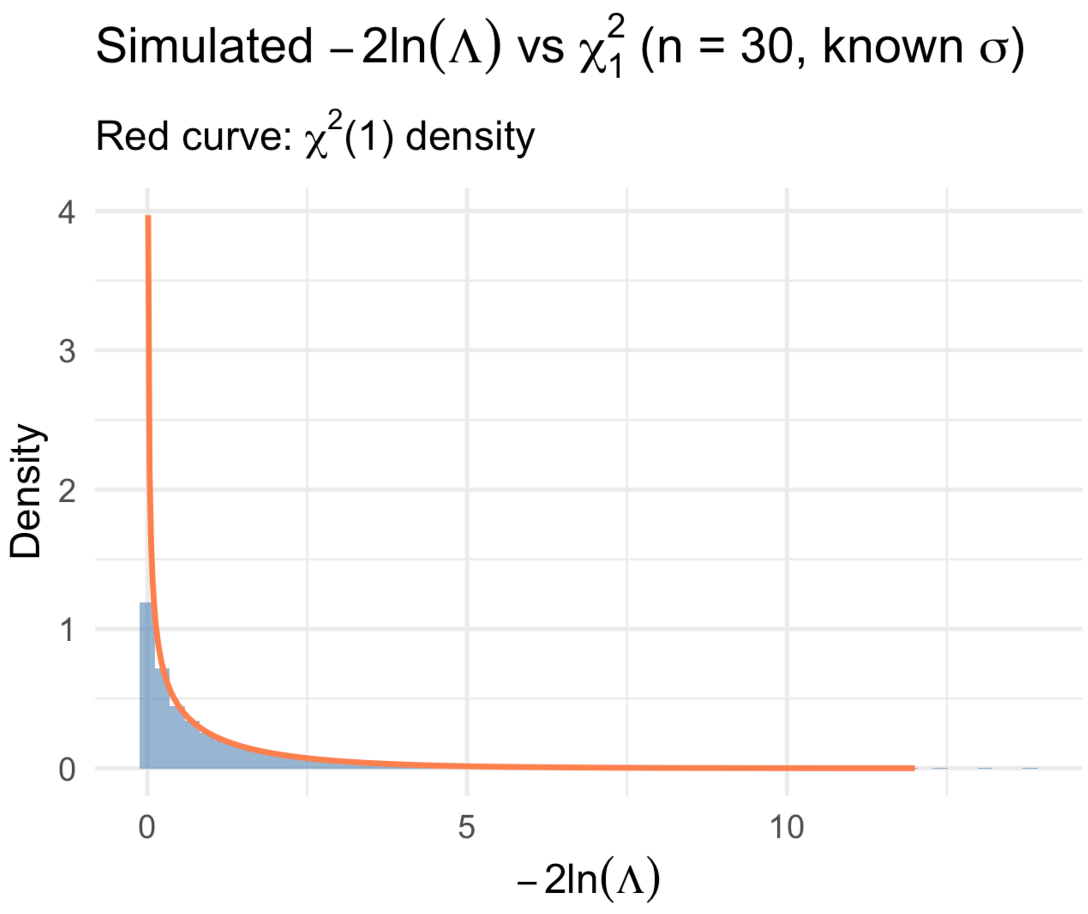
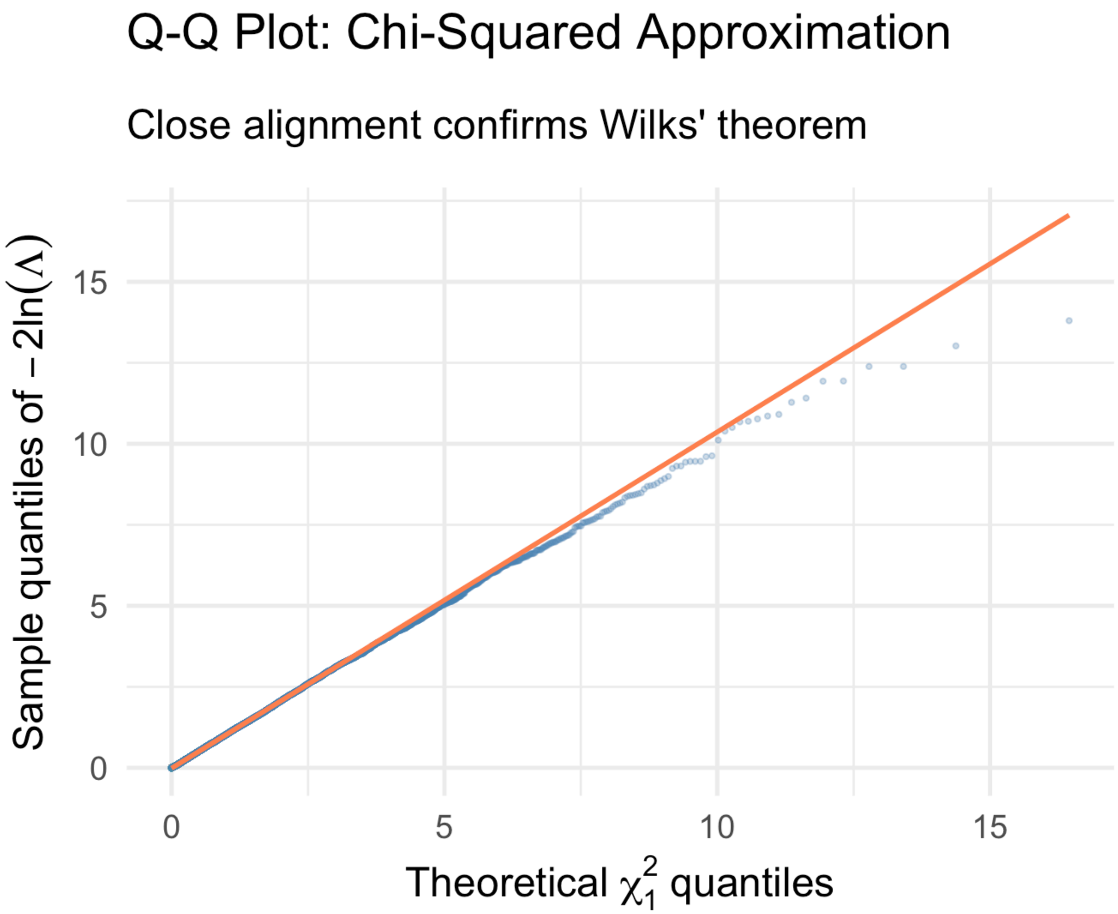
Degrees of freedom: $m = 1 - 0 = 1$, so $-2\ln(\Lambda) \mathrel{\dot\sim} \chi^2_1$ under H_0.

Reject H_0 when $-2\ln(\Lambda) \geq \chi^2_{\alpha,1}$.

Simulation: Verifying the Chi-Squared Approximation

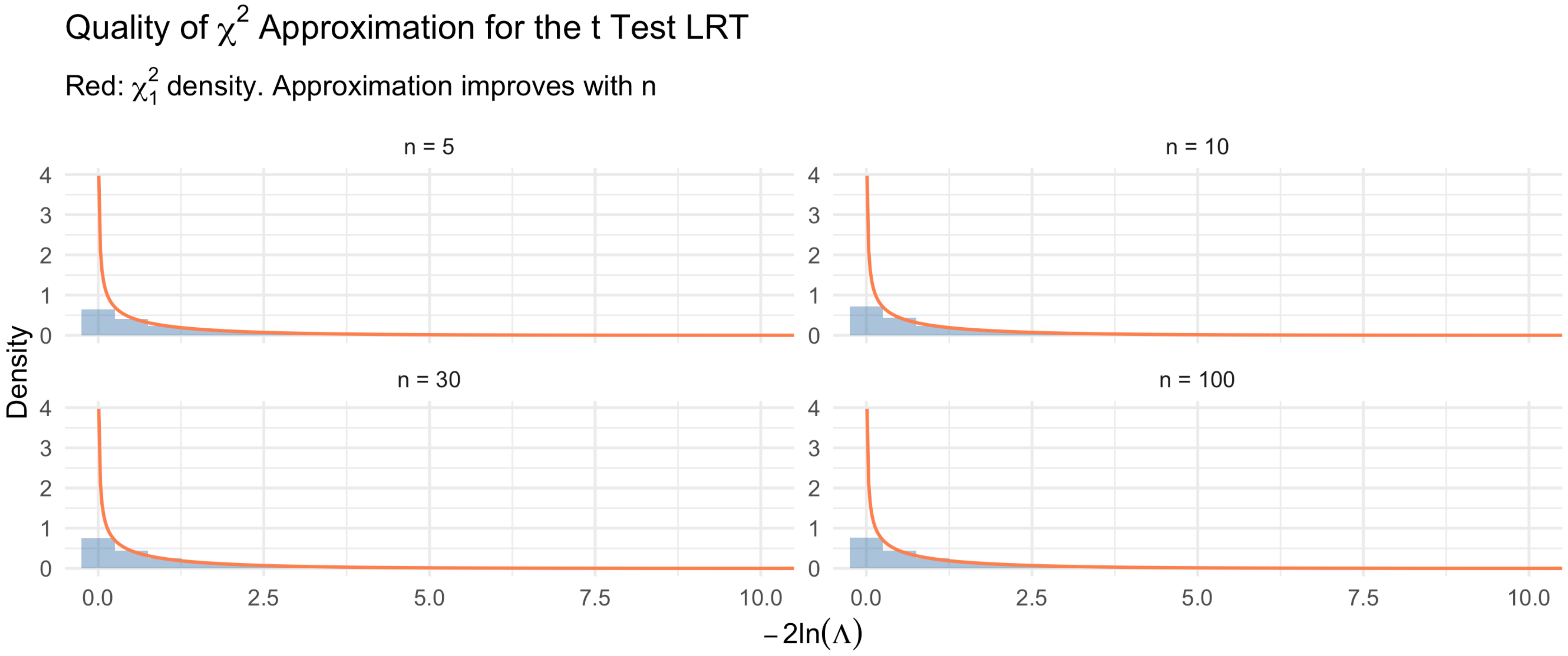
▶ Code





How Good is the Approximation for Small n?

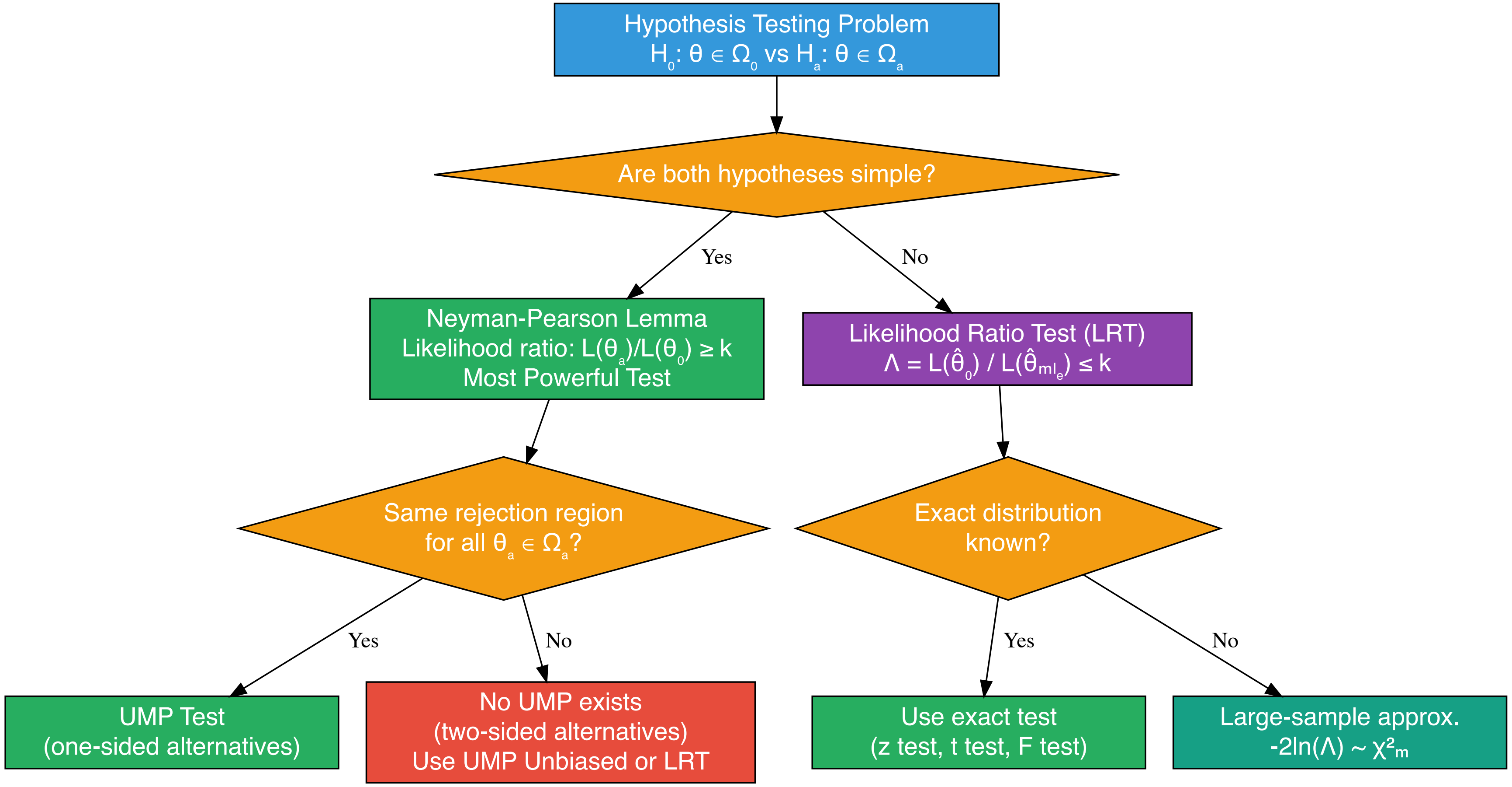
▶ Code



Connecting the Pieces

The Big Picture: Hypothesis Testing Framework

▶ Code



Summary: Classical Tests as LRTs

Test	Hypotheses	LRT Produces	Exact or Approx?
One-sample z	$H_0 : \mu = \mu_0, \sigma$ known	$z = \frac{\bar{x}-\mu_0}{\sigma/\sqrt{n}}$	Exact
One-sample t	$H_0 : \mu = \mu_0, \sigma$ unknown	$t = \frac{\bar{x}-\mu_0}{s/\sqrt{n}}$	Exact
Pooled two-sample t	$H_0 : \mu_1 = \mu_2$, equal σ	Pooled t statistic	Exact
Chi-squared goodness-of-fit	$H_0 : p_i = p_{i,0}$	$\sum(O_i - E_i)^2/E_i$	Large-sample
F test (ANOVA)	$H_0 : \mu_1 = \cdots = \mu_k$	F ratio	Exact (normal)

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Key Insight

The LRT principle is a **unifying framework**. Many of the “named” tests from introductory statistics — z tests, t tests, F tests, chi-squared tests — can all be derived as likelihood ratio tests. The LRT gives us a principled way to construct new tests for situations not covered by standard formulas.

Comparison of Approaches: When to Use What

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- Applies to **simple vs simple** hypotheses
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Likelihood Ratio Tests:

- Apply to **any** testing problem (simple or composite, one or multiple parameters)
- **General-purpose recipe** that always yields a reasonable test
- Often coincide with optimal tests when those exist
- Large-sample χ^2 approximation makes them practical even when exact distributions are unknown

Key Takeaways and Looking Ahead

Key Takeaways

Today's Main Points

1. **Simple hypotheses** completely specify the distribution; composite hypotheses do not
2. **Neyman-Pearson Lemma:** the likelihood ratio test $L(\theta_a)/L(\theta_0) \geq k$ is the most powerful test for simple H_0 vs simple H_a
3. **Power function** $\pi(\theta') = P_{\theta'}(\text{Reject } H_0)$ characterizes test performance across all parameter values
4. **UMP tests** maximize power for *every* alternative; they exist for one-sided tests in exponential families
5. **Likelihood ratio test** $\Lambda = L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ provides a general recipe for composite hypotheses
6. **LRT $\Rightarrow$ classical tests:** the z test and t test are both LRTs
7. **Wilks' theorem:** $-2\ln(\Lambda) \xrightarrow{d} \chi^2_m$ provides a large-sample approximation

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Common mistakes to avoid:

- Confusing the NP ratio $L(\theta_a)/L(\theta_0)$ (reject when large) with the LRT ratio $L(\hat{\theta}_0)/L(\hat{\theta}_{mle})$ (reject when small)
- Assuming UMP tests always exist (they don't for two-sided alternatives)
- Using the χ^2 approximation when n is too small
- Forgetting that the LRT degrees of freedom m count *free* parameters, not total parameters

Looking Ahead

Next class (Lesson 16): Further Aspects of Hypothesis Testing; Bootstrap Hypothesis Testing (Devore 9.6, C&H Chapter 8)

- Statistical vs practical significance
- Relationship between confidence intervals and hypothesis tests
- Bootstrap hypothesis testing
- Permutation tests

References

- Devore, Berk, and Carlton. *Modern Mathematical Statistics with Applications* (Springer). Section 9.5
- Chihara and Hesterberg. *Mathematical Statistics with Resampling and R* (Wiley). Section 8.4
- Neyman, J. & Pearson, E. S. (1933). On the problem of the most efficient tests of statistical hypotheses. *Philosophical Transactions of the Royal Society A*, 231, 289–337.
- Wilks, S. S. (1938). The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1), 60–62.

Questions?

Thank you!